

Integration by parts.

$\nwarrow$ inv. trig
 $\nwarrow$ A T E $\nwarrow$ Expo.
 $\uparrow$ log $\uparrow$ Trig
 $\uparrow$ algebraic

$$\int \ln(x) dx$$

$$\int \ln^2(x) dx$$

$$\textcircled{4} \int y e^{y-2}$$

$$\textcircled{1} \int x^2 e^{3x} dx$$

$$\textcircled{2} \int \arcsin(x) dx$$

$$\textcircled{3} \int e^{2x} \cos(3x) dx$$

$\textcircled{2}$
 D

$u = \arcsin(x)$
 $du = \frac{1}{\sqrt{1-x^2}} dx$

$v = x$
 $dv = dx$

$\nearrow I$

$$\underbrace{\int u dv}_{I} = uv - \int v du.$$

$$I = x \arcsin(x) - \int \frac{x}{\sqrt{1-x^2}} dx$$

$$u = 1-x^2$$

$$du = -2x dx \Rightarrow x dx = -\frac{1}{2} du$$

$$I = x \arcsin(x) + \frac{1}{2} \int \frac{1}{\sqrt{u}} du = x \arcsin(x) + \sqrt{1-x^2} + C$$

$$2u^{1/2} + C = 2(1-x^2)^{1/2} + C$$

$$\textcircled{1} \int x^2 e^{3x} dx$$

$$\begin{array}{l} u = x^2 \quad v = \frac{1}{3} e^{3x} \\ \downarrow \quad \quad \quad \nearrow \\ du = 2x dx \quad dv = e^{3x} dx \end{array}$$

$$I = \frac{1}{3} x^2 e^{3x} - \frac{2}{3} \int \underbrace{x}_u \underbrace{e^{3x} dx}_{dv}$$

$$\begin{array}{l} u = x \quad v = \frac{1}{3} e^{3x} \\ \downarrow \quad \quad \quad \nearrow \\ du = dx \quad dv = e^{3x} dx \end{array}$$

$$I = \frac{1}{3} x^2 e^{3x} - \frac{2}{3} \left(\frac{1}{3} x e^{3x} - \frac{1}{3} \int e^{3x} dx \right)$$

$$= \frac{1}{3} x^2 e^{3x} - \frac{2}{9} x e^{3x} + \frac{2}{9} \underbrace{\int e^{3x} dx}_{\frac{1}{3} e^{3x}}$$

$$I = \frac{1}{3} x^2 e^{3x} - \frac{2}{9} x e^{3x} + \frac{2}{27} e^{3x} + C$$

Antiderivatives of $x^2 e^{3x}$.

$$\textcircled{3} \int e^{2x} \cos(3x) dx$$

$$u = \cos(3x) \quad \text{new } v = \frac{1}{2} e^{2x}$$

~~old~~

$$du = -3 \sin(3x) dx \quad dv = e^{2x} dx$$

$$I = \frac{1}{2} e^{2x} \cos(3x) + \frac{3}{2} \int e^{2x} \underline{\sin(3x)} dx$$

$$\begin{array}{l} u = \sin(3x) \quad v = \frac{1}{2} e^{2x} \\ \downarrow \quad \quad \quad \downarrow \\ du = 3 \cos(3x) dx \quad dv = e^{2x} dx \end{array}$$

$$I = \frac{1}{2} e^{2x} \cos(3x) + \frac{3}{2} \left(\frac{1}{2} e^{2x} \sin(3x) - \underbrace{\frac{3}{2} \int e^{2x} \cos(3x) dx}_I \right)$$

$$I = \frac{1}{2} e^{2x} \cos(3x) + \frac{3}{4} e^{2x} \sin(3x) - \frac{9}{4} I$$

$$+\frac{9}{4} I$$

$$+\frac{9}{4} I$$

$$I + \frac{9}{4} I = \frac{1}{2} e^{2x} \cos(3x) + \frac{3}{4} e^{2x} \sin(3x)$$

$$\frac{13}{4} I = \frac{1}{2} e^{2x} \cos(3x) + \frac{3}{4} e^{2x} \sin(3x)$$

$$\cdot \frac{4}{13}$$

$$I = \frac{2}{13} e^{2x} \cos(3x) + \frac{3}{13} e^{2x} \sin(3x) + \underline{\underline{C}}$$

$$\int_a^b u dv = uv \Big|_a^b - \int_a^b v du$$

$$\int u dv = \underline{uv} - \int \underline{v du}$$

$$\begin{cases} F'(x) = f(x) \\ \int f(x) dx = F(x) + C \end{cases}$$

$$\int x f(x) dx = x F(x) - \int F(x) dx$$

$$\textcircled{1} \begin{cases} u = x & v = F(x) \\ \downarrow du = dx & \downarrow dv = f(x) dx \end{cases}$$

$$\textcircled{2} u = f(x) \quad v = \int dv = \int x dx = \frac{1}{2} x^2$$

$$du = f'(x) dx \quad dv = x dx$$

$$\int x f(x) dx = \frac{1}{2} x^2 f(x) - \int \frac{1}{2} x^2 f'(x) dx$$

$$\int f(x) e^{2x} dx = \frac{1}{2} f(x) e^{2x} - \frac{1}{2} \int f'(x) e^{2x} dx$$

$$\frac{d}{dx} \left(\frac{1}{2} f(x) e^{2x} - \frac{1}{2} \int f'(x) e^{2x} dx \right) = f(x) e^{2x}$$

$$\cancel{\frac{1}{2} f'(x) e^{2x}} + \frac{1}{2} f(x) \cdot 2e^{2x} - \cancel{\frac{1}{2} f'(x) e^{2x}} = f(x) e^{2x}$$

Trigonometric integrals - $\int \sin^m(x) \cos^n(x) dx$

1) m is odd
2) n is odd

3) both are even

$$\sin^2(x) = 1 - \cos^2(x)$$

$$\begin{aligned} \int \sin^3(x) \cos^2(x) dx &= \int \sin^2(x) \cos^2(x) \underbrace{(\sin(x) dx)}_{\text{derivative of cosine}} \\ &= \int (1 - \overbrace{\cos^2(x)}^{u^2}) \overbrace{\cos^2(x)}^{u^2} \underbrace{(\sin(x) dx)}_{-du} \end{aligned}$$

$$u = \cos(x)$$

$$du = -\sin(x) dx$$

$$= \int (1 - u^2) \cdot u^2 (-du) = - \int (1 - u^2) u^2 du$$

$$= - \int (u^2 - u^4) du = - \left(\frac{1}{3} u^3 - \frac{1}{5} u^5 \right) + C$$

$$= - \frac{1}{3} (\cos(x))^3 + \frac{1}{5} (\cos(x))^5 + C.$$

$$\int_0^{\pi/2} \sin^3(x) \underline{\underline{\cos(x) dx}}$$

$$\cos(x) dx$$

$$u = \sin(x)$$

$$du = \cos(x) dx$$

$$\int_0^1 u^3 du = \frac{1}{4} u^4 \Big|_0^1 = \frac{1}{4}$$

$$\int_0^{\pi/2} \underbrace{\sin^2(x)} \cos(x) (\sin(x) dx)$$

$$\int_0^{\pi/2} (1 - \cos^2(x)) \cos(x) \sin(x) dx$$

$$u = \cos(x)$$

$$du = -\sin(x) dx$$

$$= - \int_1^0 (1 - u^2) u du$$

$$= \int_0^1 (1 - u^2) u du$$

$$\int \sin^2(x) \cos^2(x) dx$$

$$\sin^2(x) = \frac{1 - \cos(2x)}{2}$$

$$\cos^2(x) = \frac{1 + \cos(2x)}{2}$$

$$\int \left(\frac{1 - \cos(2x)}{2} \right) \left(\frac{1 + \cos(2x)}{2} \right) dx$$

$$(1-x)(1+x) = 1-x^2$$

$$= \frac{1}{4} \int (1 + \cancel{\cos(2x)} - \cancel{\cos(2x)} - \cos^2(2x)) dx$$

$$= \frac{1}{4} \int (1 - \cos^2(2x)) dx$$

$$= \frac{1}{4} \underbrace{\int dx}_x - \frac{1}{4} \underbrace{\int \cos^2(2x) dx}_{\cos^2(2x)}$$

$$\cos^2(2x) = \frac{1 + \cos(2(2x))}{2}$$

$$= \frac{1}{4} x - \frac{1}{4} \int \frac{1 + \cos(4x)}{2} dx$$

$$\begin{aligned} \int f(x) dx &= F(x) + C \\ \int f(ax) dx &= \frac{1}{a} F(ax) + C \end{aligned}$$

$$= \frac{1}{4} x - \frac{1}{8} \int (1 + \cos(4x)) dx$$

$$\begin{aligned} &= \frac{1}{4} x - \frac{1}{8} \int dx - \frac{1}{8} \int \cos(\overset{u=4x}{4x}) dx = \frac{1}{4} x - \frac{1}{8} x - \frac{1}{32} \int \cos(u) du \\ &= \frac{1}{4} x - \frac{1}{8} x - \frac{1}{32} \sin(4x) + C. \end{aligned}$$

$du = 4 dx$
 $\frac{1}{4} du = dx$